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Hedging and boosting in abstracts of applied linguistics articles: A comparative study of English- and Chinese-medium journals

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ABSTRACT

Hedges and boosters are important metadiscursive resources for writers to mark their epistemic stance and position writer–reader relations. Building on previous research that suggests notable cross-cultural and cross-linguistic differences in the use of hedges and boosters in academic discourse, this comparative study investigates the use of such discourse markers in academic article abstracts. Based on a corpus of 649 abstracts collected from 8 journals of applied linguistics, this study examines if hedging and boosting strategies differ (a) between applied linguists publishing in Chinese- and English-medium journals and (b) between authors of empirical and non-empirical academic articles. Quantitative analyses indicated that abstracts published in English-medium journals featured markedly more hedges than those published in Chinese-medium journals and that abstracts of empirical research articles used significantly more boosters than those of non-empirical academic articles. Textual analyses further revealed that the distinct patterning of hedges and boosters in Chinese and English abstracts had a joint, interactive effect on the authorial certainty and confidence conveyed therein. These results are discussed in terms of culturally preferred rhetorical strategies, epistemological beliefs, lack of facility in English as a second/foreign language, and the nature of supporting evidence drawn on for knowledge claims in different types of academic writing.

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1. Introduction

This study examines the use of hedges and boosters as metadiscourse markers in the genre of the academic article abstract from a comparative perspective. According to an often-quoted definition, metadiscourse consists of "the self-reflective expressions used to negotiate interactional meanings in a text, assisting the writer (or speaker) to express a viewpoint and engage with readers as members of a particular community" (Hyland, 2005a:37). As a repertoire of rhetorical resources deployed to manage the relations among the writer, the evolving text and the intended reader, metadiscourse has attracted increasing research attention in the past decade, especially from researchers of scientific and scholarly writing (e.g., Abdi, 2002; Abdi et al., 2010; Ädel, 2006; Dahl, 2004; Gillaerts and Van de Velde, 2010; Hyland, 2005a,b; Hyland and Tse, 2004; Lindeberg, 2004; Peterlin, 2005). This focus on metadiscourse has been motivated by the growing recognition that academic writers do not simply report their findings in an objective or impersonal way, but actively draw from a range of rhetorical strategies rooted in their own disciplines and socio-cultural milieus to organize arguments, provide evidence, and evaluate claims to convince their readers (Abdi et al., 2010; Bazerman, 1988; Crismore et al., 1993; Flowerdew, 1997;

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Hyland, 2005a; Vande Kopple, 1985). Metadiscursive resources are indispensable to such social and interpersonal engagements. This is most evident in the useful distinction between interactive and interactional metadiscourse drawn by Hyland and Tse (2004). While interactive metadiscourse organizes texts and guides readers through them, interactional metadiscourse engages and orients readers towards writers' perspectives on propositional content, their imagined audience and themselves (Hyland, 2005a; Hyland and Tse, 2004; Thompson and Thetela, 1995). Thus, in academic texts such as research articles, metadiscourse plays a key role in knowledge construction through managing the interactions between writers and readers who often come from the same discourse community with shared cultural, academic, and rhetorical practices. The strategic use of metadiscourse in academic writing not only increases the chances of knowledge claims being accepted, but also indexes a writer's competence as a participant in the discourse community involved (Hyland, 2005a).

Hedges and boosters constitute two categories of interactional metadiscourse strategies that are frequently employed in academic writing, particularly in the genre of the research article (Hyland, 2005b; Hyland and Tse, 2004). Hedges are self-reflective linguistic expressions (e.g., *might*, *suggest*, *probably*) employed to express epistemic modality and modify the illocutionary force of speech acts (Holmes, 1982, 1988). They can be used to qualify the writer's commitment to a proposition (Vande Kopple, 1985), to show uncertainty about the truth of an assertion (Crismore et al., 1993), to "withhold commitment and open dialogue" (Hyland, 2005a:49) by acknowledging alternative viewpoints or the subjectivity of one's own position, and/or "to mitigate the force of an utterance 'for the sake of politeness'" (Holmes, 1990:185). Boosters, on the other hand, are linguistic devices (e.g., *demonstrate*, *undoubtedly*, *it is clear that*) that increase the illocutionary force of speech acts (Holmes, 1984), emphasize certainty about a proposition or confidence in an assertion (Abdi et al., 2010; Holmes, 1988), express authorial commitment to a proposition (Crismore et al., 1993; Millan, 2008) or close off alternative viewpoints by strengthening the asserted position (Gillaerts and Van de Velde, 2010; Hyland, 1998a). As Hyland (2005a) puts it, "by closing down possible alternatives, boosters emphasize certainty and construct rapport by marking involvement with the topic and solidarity with an audience, taking a joint position against other voices" (p. 53). Thus, hedges and boosters are, in a sense, two sides of the same coin: they are metadiscursive resources that the writer can capitalize on to express uncertainty or certainty about a proposition, withhold or strengthen commitment to a position, entertain or dismiss alternatives, open or close dialogue with the reader, and attenuate or boost illocutionary force (Holmes, 1984; Millan, 2008). The skillful manipulation of hedges and boosters in academic texts not only signals a writer's epistemic stance towards propositional content and intended readers, but also marks him/herself as a competent member of the discourse community (Hyland, 1998a, 2005a).

It is clear from the characterization above that, distinct from and yet closely related to propositional discourse, metadiscourse allows writers to construct and organize their text in anticipation of the intended readers' knowledge, interests and expectations, and to project themselves into their text to manage interaction with the readers and influence their reactions (Dahl, 2004; Millan, 2008). Metadiscourse as such is likely to reflect "cultural norms, values, and belief systems prevailing in discourse communities which constitute social contexts of texts" (Golebiowski, 2002:59). As hedges and boosters are used by writers to "engage with the socially determined positions of others" (Hyland, 2005a:52) and to mark commitments, beliefs and attitudes which are often socioculturally situated, there is good reason to expect their use to vary across different cultural and language communities with their own communicative norms, discursive practices, rhetorical conventions, and power relations (Connor, 1996; Holmes, 1988).

This expectation has been largely supported by a growing body of research that has investigated the use of these metadiscourse strategies by members of different language, cultural, or disciplinary communities. Several studies (Fløttum et al., 2006; Kong, 2006; Martín-Martín, 2008; Martín-Martín and Burgess, 2004; Salager-Meyer et al., 2003; Vassileva, 2001; Vold, 2006b) have compared the use of hedges and/or boosters by academic writers from different culture/language groups and found cross-cultural/linguistic differences in the use of these metadiscourse markers to negotiate knowledge claims and construct scientific knowledge. Vassileva (2001), for instance, compared English research articles by Anglo-American linguists with Bulgarian and English research articles by Bulgarian linguists and identified considerable differences among the three sets of articles in the use of hedges and boosters to convey different degrees of commitment/detachment. In a similar vein, Vold (2006b) examined the use of selected epistemic modality markers as hedging devices in a corpus of 120 research articles written in English, French and Norwegian, and found evident cross-linguistic differences. Specifically, the English and Norwegian research articles used significantly more hedges than the French research articles did. Language- and culture-specific differences in using hedges were also observed in Kong's (2006) study on how other people's ideas were evaluated in 80 research articles written in Chinese and English in the humanities, social sciences, and hard sciences. Another group of studies (e.g., Abdi, 2002; Hyland, 1998a,b, 2005b; Millan, 2008; Vassileva, 2001; Vold, 2006a) examined the use of hedges and boosters across disciplines and found evidence of disciplinary influences. For example, Hyland (1998a) investigated the use of hedges and boosters in a corpus of 56 research articles from eight disciplines and found considerable differences in the use of these metadiscursive resources between the natural sciences on the one hand, and the humanities and social sciences on the other. Similarly, Abdi (2002) examined the use of interpersonal metadiscourse markers, including hedges and boosters, in 55 research articles from the natural and social sciences and found significant interdisciplinary differences in the use of hedges but no differences in the use of boosters. In a recent study based on a larger corpus of 240 research articles from the same eight disciplines, Hyland (2005b) corroborated his earlier finding that there were appreciable differences in the use of hedges and boosters between "soft" (social sciences and the humanities) and "hard" (natural sciences) disciplines. Clear disciplinary differences in the use of hedges and boosters were also identified in Millan's (2008) study of 96 research articles in two soft and two hard disciplines.

Despite the substantial number of empirical investigations into hedges and boosters as metadiscursive resources in academic writing, there are several issues that merit further research. To begin with, it is not clear to what extent observed differences in the use of these metadiscourse strategies are attributable to cultural and disciplinary influences, respectively (Martín-Martín and Burgess, 2004). In a study of endophorics (i.e., expressions referring to other parts of the text; see Hyland, 2005a) and frame markers in academic writing, Dahl (2004) analyzed 180 published research articles in three languages (i.e., English, French, and Norwegian) and three disciplines (i.e., economics, linguistics, and medicine). The researcher found marked differences in the use of the target metadiscourse markers among the nine subcorpora of research articles. More importantly, whether the observed differences were attributable to cultural or disciplinary influences depended on the discipline in question. There is good reason to expect a similar complex pattern of cultural and disciplinary influences to impact on the use of hedges and boosters in academic discourse. This calls for further research focused on identifying differences in the use of hedges and boosters that can be unambiguously attributed to cultural influences, disciplinary norms, and/or an interaction of the two, respectively.

A second issue that needs to be addressed is a conspicuous lack of research into the use of metadiscourse, including hedges and boosters, in relation to different types of academic writing. Previous research on academic discourse has divided the genre of the journal article into various subtypes. For example, Tarone et al. (1998) distinguished between reports of experimental/empirical studies and argumentation papers; Salager-Meyer (1992) made a three-way distinction between research articles, case reports, and review articles; and Swales (2004) subcategorized the genre of research articles into theoretical articles, review articles, data-based research articles, and shorter communications (also see Bazerman, 1988). For the purposes of this study, these different text types can be grouped, on the basis of their primary purposes and their relation to empirical data, into two broad genres of scientific writing: empirical research articles (i.e., reports of empirical studies) and non-empirical academic articles (i.e., review articles, theoretical articles, methodological articles, etc.). The former usually take "a highly codified, institutionalized form" (Bazerman, 1988:259), whereas the latter typically adopt a more fluid, purpose-specific structure (American Psychological Association, 2010; Swales, 2004). The different nature of these two different genres of academic writing may very well find expression in the metadiscourse strategies employed by authors to facilitate the knowledge construction process (Dahl, 2004). Working with an institutionalized text format (i.e., the Introduction–Method–Results–Discussion structure), authors of empirical research articles are expected to report on empirical results and assume an author-as-researcher identity (Fløttum, 2003). By contrast, no such well established text format exists for non-empirical academic articles, whose authors will have to negotiate their own text patterns and, in the words of Dahl (2004:1807), "create their findings through argumentation." As a result, authors of non-empirical academic articles are more likely to assume an author-as-acting-agent identity (Fløttum, 2003) and tend to be more visible in their writing than authors of empirical research articles. These differences in the nature and textual structure of empirical and non-empirical academic articles as well as in the corresponding identity assumed by their authors are likely to create different demands for authorial management of interaction with the imagined readers and manipulation of their reactions through metadiscourse strategies such as hedges and boosters. Consequently, a comparative study of these metadiscourse markers in empirical and non-empirical academic articles can be expected to yield important insights into how they are deployed to facilitate knowledge construction in academic discourse (Pho, 2008).

The growing body of research on hedges and boosters notwithstanding, little of this research has focused on the genre of the article abstract. This lack of attention is unexpected in view of the recent insights into the pragmatic functions, linguistic features, and rhetorical structure of abstracts (Dahl, 2009; Lindeberg, 2004; Stotesbury, 2003). As a constitutive part of an academic article, the abstract not only provides an instructive summary of the accompanying article (Bhatia, 1993; Swales, 1990), but also serves an advertising function to promote it (Berkenkotter and Huckin, 1995; Dahl, 2009; Hyland and Tse, 2005a,b; Lindeberg, 2004). Furthermore, as Gillaerts and Van de Velde (2010) point out, "from an embedded genre. . .functionally and structurally placed in between the title and (the introduction of) the [research article] and as a rule to be read in combination with the full-text article, [abstracts] are gradually changing into a stand-alone genre, often consulted on their own by scholars who want to assess the RA's relevance for the field" (p. 136). As an academic genre that is gaining prominence and importance (Berkenkotter and Huckin, 1995; Lindeberg, 2004), the abstract is, by nature, a social response intended to orient readers in navigating through the exponentially growing output of academic publications, and is hence "replete with subjective material, expressed by interactional elements" (Gillaerts and Van de Velde, 2010:130). Thus, it is an ideal genre for studying how metadiscursive resources such as hedges and boosters are drawn on to mark knowledge claims and orient prospective readers. Yet, despite a sustained interest in its rhetorical organizations (e.g., Lorés, 2004; Martín-Martín, 2003; Pho, 2008; Samraj, 2005; Santos, 1996; Stotesbury, 2003), little research has examined the use of hedges and boosters in this academic genre, not to mention doing so from a comparative perspective. One notable exception is Lindeberg's (2004) study of how boosters and hedgers were used as strategies of promotion and mitigation in the abstracts as well as other sections of journal articles from three business disciplines. Two other exceptions are Gillaerts and Van de Velde (2010), which examined, among other things, the use of hedges and boosters in research article abstracts from a historical perspective, and Martín-Martín and Burgess (2004), which investigated the use of various rhetorical strategies, including hedges, to convey academic criticism in English and Spanish research article abstracts in two social sciences.

In response to the issues identified above, this study aims to investigate cultural, language, and genre-based influences on the use of hedges and boosters by comparing abstracts accompanying empirical vs. non-empirical academic articles published in leading Chinese and English journals of applied linguistics (see section 2.1 for criteria for selecting the journals). Specifically, it seeks to answer the following two research questions:

(1) Are there any differences/similarities in the use of hedges and boosters between abstracts published in major Chinese- and English-medium academic journals in the discipline of applied linguistics?
(2) Are there any differences/similarities in the use of hedges and boosters between abstracts accompanying empirical and non-empirical articles?

2. Methodology

To address the above research questions, we collected academic article abstracts published in the discipline of applied linguistics with a special emphasis on second and foreign language learning and teaching. The selection of this discipline was motivated by several considerations. First, to the best of our knowledge, very few contrastive studies of abstracts in English and Chinese have examined this discipline, leaving an obvious lacuna to fill in. Second, it would be necessary to focus on a single discipline to allow the teasing apart of cultural/language effects from disciplinary ones on the use of hedges and boosters. That is, holding the discipline variable constant would allow us to rule out variations in the disciplinary norms of metadiscursive practices as an explanation for any observed systematic difference in the use of hedges and boosters. Third, as applied linguists who regularly read academic papers published in leading English- and Chinese-medium journals, we had an insider knowledge of the discourse practices of the sub-communities concerned. This intimate knowledge would be an advantage in data coding and analysis.

In designing our study, we followed a methodological framework proposed by Moreno and colleagues (Connor and Moreno, 2005; Moreno and Suárez, 2008) for identifying recurrent differences in the use of rhetorical resources in academic texts across languages and cultures. A central notion of this framework is *tertium comparationis* or equivalence (Connor and Moreno, 2005). According to the methodological framework, a common platform of comparison should be established to make sure that "we compare elements that can in fact be compared" (p. 154). To this end, equivalence should be achieved at a variety of levels, including the selection of textual data, the identification of target conceptual categories, and the formulation of an explicit taxonomy for data coding and analysis. It is equivalence at these various levels that allows the researcher to make meaningful comparisons of quantitative results for each target rhetorical category and draw reliable and valid conclusions about cross-linguistic/cultural similarities and differences.

2.1. Construction of equivalent subcorpora

To construct parallel subcorpora of abstracts, we made every effort to select comparable English- and Chinese-medium journals displaying maximum similarity. The selection of the source journals was based on three sources of information: (a) the Social Sciences Citation Index (ISI Web of Science, 2008); (b) the Chinese Social Sciences Citation Index (Chinese Social Sciences Research Evaluation Center, 2008); and (c) previous research on applied linguistics journals (i.e., Benson et al., 2009; Gao et al., 2001; Shi et al., 2005). All the selected journals were included in one of the two influential citation indices and more than one of the previous studies. The Chinese-medium journals were (a) *Foreign Language Teaching and Research* (*FLTR*), (b) *Foreign Language World* (*FLW*), (c) *Journal of Foreign Languages* (*JFL*), and (d) *Modern Foreign Languages* (*MFL*). All four were refereed journals published in Mainland China. The English-medium journals were (a) *Applied Linguistics* (*AP*), (b) *Modern Language Journal* (*MLJ*), (c) *Studies in Second Language Acquisition* (*SSLA*), and (d) *TESOL Quarterly* (*TQ*). All four were international refereed journals published in the UK or the USA.

The two groups of journals were highly comparable in major respects. First, as evidenced in their inclusion in the two influential citation indices and the published studies, both groups were the leading academic journals in the two languages, were widely read by their respective communities of scholars, and represented the best of applied linguistics research in China and internationally. Second, in terms of genre and subject matter, both groups published empirical and non-empirical academic articles in the areas of second or foreign language teaching and learning. Third, all the abstracts were taken from the publication period of 2007–2008. Because the Chinese journals had more issues each year and carried more articles in each issue, only those abstracts published in 2008 were included, whereas all the abstracts published in the English journals in 2007 and 2008 were sampled. This sampling decision was made in order to have a comparable number of abstracts from the two sets of journals.

Since the Chinese journals carried abstracts written in both Chinese and English for each article, three parallel subcorpora were constructed: (a) the CA-CJ subcorpus (i.e., Chinese abstracts published in the Chinese journals), (b) the EA-CJ subcorpus (i.e., English abstracts published in the Chinese journals), and (c) the EA-EJ subcorpus (i.e., English abstracts published in the English journals). Table 1 summarizes descriptive information about the three subcorpora.

A biographical search revealed that 19 of the 195 abstracts in the EA-EJ subcorpus were authored or co-authored by Chinese applied linguists. In case the Chinese applied linguists might have transferred Chinese rhetorical practices into their writings in English and skewed the pattern of use of hedges and boosters in the EA-EJ subcorpus, preliminary analyses were conducted to compare the 19 abstracts with those authored by non-Chinese applied linguists. An independent-samples *t* test indicated that there was no statistically significant difference in the mean number of hedges used between the abstracts authored by the Chinese applied linguists ($M = 1.58$, $SD = 1.68$) and those authored by their non-Chinese counterparts ($M = 2.09$, $SD = 2.02$), $t(193) = 1.05$, $p = .22$. Neither was there a statistically significant difference in the use of boosters between the Chinese applied linguists ($M = 0.74$, $SD = 0.65$) and the non-Chinese applied linguists ($M = 0.90$, $SD = 1.09$), $t(193) = .63$, $p = .21$. These results were consistent with Lindeberg's (2004) contention that the stringent quality demands of top English-medium research journals and their rigorous review processes "make it irrelevant whether the RAs were written

Table 1
Profile of the parallel subcorpora.

Subcorpus	No. of abstracts	No. of English words/Chinese characters	Mean length (English words/Chinese characters)
CA-CJ	227	45,715	201.39
FLTR	53	10,479	197.72
FLW	77	13,613	176.79
JFL	54	11,216	207.70
MFL	43	10,407	242.02
EA-CJ	227	24,538	108.10
FLTR	53	5948	112.23
FLW	77	7016	91.12
JFL	54	6018	111.44
MFL	43	5556	129.21
EA-EJ	195	33,213	170.32
AL	59	9842	166.81
MLJ	63	10,631	168.75
SSLA	32	5759	179.97
TQ	41	6981	170.27

by native English speakers or not" (p. 8). Given the results, the two subgroups of English abstracts were combined for subsequent analyses.

2.2. Operationalization of hedges and boosters

Based on Hyland's metadiscourse model (see Hyland, 2005a; Hyland and Tse, 2004) and Holmes's research on linguistic resources for expressing epistemic and affective meaning in English (see Holmes, 1982, 1988, 1990), we identified a number of defining features of hedges and boosters in order to facilitate the data coding process: (a) hedges and boosters are metadiscourse markers explicitly used in a text; (b) hedges and boosters signal writers' stance over entire propositions rather than modify individual lexis; and (c) hedges and boosters not only communicate writers' epistemic stance but also express affective meaning by modifying the illocutionary force of speech acts. Importantly, we shared Lakoff's (1973) argument that "any adequate treatment [of hedges and, for that matter, boosters] will have to take context into account" (p. 484). As Holmes (1990) also points out in her discussion about the identification of linguistic forms as hedges and boosters, "form alone..., though it is an important first stage in analysis, is rarely a sufficient basis for categorization" (p. 188). The importance of context in determining whether specific linguistic forms are hedges or boosters can be appreciated in consideration of the distinction between the epistemic and root meanings of certain lexical verbs (Holmes, 1988), as exemplified in the uses of *suggest* in the following examples:

(1) These findings *suggest* that development of pragmatic knowledge and processing capacity of using the knowledge may not coincide perfectly in L2 development. (*TQ*7, EA-EJ)

(2) In the concluding discussion, we offer some possible directions for extending this type of investigation and also *suggest* some practical ways in which our analysis might enrich the teaching and learning of foreign languages. (*MLJ*33, EA-EJ)

The context of (1) makes it clear that *suggest* expresses epistemic modality and is a hedge, whereas the same linguistic form in the context of (2) adds only propositional information, communicating a non-epistemic meaning.

In a similar vein, context explains why certain epistemic linguistic devices can function as hedges but others as boosters, although the former do not seem to differ much from the latter in the degree of certainty they express by themselves. For example, the Chinese epistemic verb 认为does not necessarily express a markedly different level of uncertainty from that of 指出when they are considered alone. However, (3) and (4) below demonstrate that when used in their respective contexts, 认为 (i.e., *hold*) conveys tentativeness about the author's position by implicitly recognizing its subjectivity, whereas 指出 (i.e., *show*) boosts commitment to the author's claim by expressing a factive meaning:

(3) 考虑到大学英语口语考试的实际情况,本文**认为**整体评分标准是最符合大学英语大面积口试实际、操作性较强、效度和信度相对较高的评分标准。(*FLTR*53, CA-CJ)

(Considering the practicalities of oral English assessment at the college level, this paper *holds* that holistic scoring best meets the practical needs of requiring large numbers of college students to take an oral English proficiency test, is easy to operate, and has relatively high validity and reliability.)[1]

[1] All English translations of the Chinese examples are our own. The corresponding English abstracts in the EA-CJ subcorpus are not literal translations of the Chinese abstracts from which the examples are taken.

Table 2
The finalized taxonomy of English and Chinese hedges and boosters.

Metadiscourse marker	English exemplar	Chinese exemplar
Hedges		
Modal auxiliaries	Might, could, would	可能 (may)
Epistemic lexical verbs	Seem, assume, suggest	觉得 (feel), 倾向 (tend to), 尝试 (attempt)
Epistemic adjectives and adverbs	Perhaps, likely, mainly	相对 (relatively), 主要 (mainly), 比较 (kind of)
Miscellaneous[a]	In general, assumption (that)	在一定程度上 (to certain extent), 总体来说 (in general)
Boosters		
Modal auxiliaries	Must (possibility), will	会 (will), 要 (shall), 必须 (must)
Epistemic lexical verbs	Demonstrate, show, find	表明 (show), 发现 (find), 指出 (point out)
Epistemic adjectives and adverbs	Actually, always, clearly	实际上 (in fact), 明显 (obviously), 总是 (always)
Miscellaneous[a]	It is well known (that), the fact that	不可否认 (undeniably), 普遍认为 (it is well known)

[a] The miscellaneous categories comprised, among others, some common knowledge markers (Koutsantoni, 2004), modal nouns (Hyland, 1996), and the epistemic *that*-constructions (Hyland and Tse, 2005a,b).

(4) 本文则**指出**，语法隐喻理论除了对论题二显示出深刻的洞见外,对其他三个论题的论证都不能成立。(*JFL*39, CA-CJ)

(However, this paper *shows* that although it offers deep insight into Thesis Two [Halliday's] grammatical metaphor theory does not hold for the other three theses.)

A taxonomy of English and Chinese hedges and boosters was developed as a coding scheme on the basis of previous research on hedges and boosters (e.g., Crismore et al., 1993; Holmes, 1982, 1984, 1988; Hyland, 1996, 1998a,b, 2005a; Hyland and Tse, 2004; Vande Kopple, 1985). The list of frequently occurring English hedges and boosters compiled by Hyland (2005a) was a useful starting point for coding English abstracts. Since there was no existing inventory of Chinese hedges and boosters, we began by translating as many English hedges and boosters as possible into their Chinese equivalents. As Holmes (1982) points out, however, not all linguistic devices for expressing epistemic modality and modifying illocutionary force can be translated unproblematically from one language into another. Consequently, we also reviewed the relevant Chinese research literature on modality (Hsieh, 2006; Lu, 2004; Xu, 2007; Zhu, 2005), epistemic adverbs and adverbials (Zhang, 2000), pragmatic markers (Feng, 2008), and metadiscourse markers (Li, 2006) to identify additional Chinese hedges and boosters. As a result, a comparable list of Chinese hedges and boosters was compiled for coding the CA-CJ subcorpus. The two lists were then revised and refined iteratively in a preliminary coding of the data until there were no further changes to them. For the sake of space constraints, the complete lists of Chinese and English hedges and boosters that were identified in our data are not provided here. Only the finalized taxonomy, together with a few exemplars of the various categories, is presented in Table 2. Notably, the major formal categories of hedging and boosting devices in the taxonomy are essentially consistent with those in the classificatory schemes adopted by Hyland (1996, 1998b), Holmes (1982, 1988), and Millan (2008).

2.3. Procedure

The data coding was carried out in two stages: (a) the coding of hedges and boosters in the abstracts; (b) the coding of the accompanying articles into empirical and non-empirical ones. In the first stage, given the known multifunctionality of linguistic forms, all the abstracts were coded independently by one of the authors and another trained coder. A training session was conducted before the two coders used the coding scheme to code all the abstracts in two rounds. In the first round, the two coders independently coded all the data and reached an average agreement rate of approximately 80%. Based on discussion about the disagreed cases, the coding scheme was revised and applied in the second round of coding. The final inter-coder agreement rate was 95.5% for hedges and 96.2% for boosters. Again, the two coders discussed their disagreements and decided to exclude from further analyses those borderline or problematic cases on which they could not reach agreement even after discussion. The excluded data were less than 4% of the total cases.

In the second coding stage, all the abstracts were classified into two broad categories: those accompanying empirical research articles and those accompanying non-empirical academic articles. The empirical category included articles reporting on investigations that employed a quantitative, qualitative or mixed approach to collecting and analyzing primary data (Benson et al., 2009; Gao et al., 2001). The non-empirical category comprised mainly articles synthesizing and evaluating existing research, advancing theories and implications, discussing methodological issues in research, addressing pedagogical issues, or summarizing personal experience/viewpoints. We first examined all the abstracts to identify those which explicitly aligned themselves to the empirical or non-empirical category. For the abstracts which did not do so, we turned to the relevant sections of the accompanying articles to determine which category they belonged to.

Table 3
Descriptive statistics for the use of hedges and boosters by subcorpus and article type.

Metadiscourse marker	Article type	CA-CJ			EA-CJ			EA-EJ		
		n	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*
Hedges	Empirical	73	0.79	1.04	73	1.23	1.27	151	2.03	2.00
	Non-empirical	154	0.86	1.12	154	0.86	0.99	44	2.07	1.99
	Total	227	0.84	1.09	227	0.98	1.10	195	2.04	1.99
Boosters	Empirical	73	1.38	0.98	73	1.00	0.75	151	0.95	1.03
	Non-empirical	154	0.79	1.00	154	0.56	0.89	44	0.64	1.08
	Total	227	0.98	1.03	227	0.70	0.87	195	0.88	1.05

3. Results

To address our research questions, both statistical and textual analyses were conducted. We first present the results of the quantitative analyses. The alpha was set at .05 (2-tailed) for the statistical tests, with a Bonferroni correction applied where appropriate.[2] We then summarize prominent patterns identified by the qualitative analyses.

3.1. Frequencies of hedges and boosters across the subcorpora

Table 3 presents descriptive statistics for the use of hedges and boosters by subcorpus and article type. There were considerable differences in the mean frequency with which hedges were used among the three subcorpora and between the two article types. For the abstracts of the empirical research articles, the mean frequency for the EA-EJ subcorpus far surpassed that for the CA-CJ subcorpus, with the value of the EA-CJ subcorpus falling in between. With regard to the abstracts of the non-empirical academic articles, the mean frequency for the EA-EJ subcorpus was markedly higher than the respective values of the CA-CJ and EA-CJ subcorpora. An almost reverse pattern was found for the mean frequency with which boosters were used. For the abstracts of both the empirical and non-empirical academic articles, the CA-CJ subcorpus had an appreciably higher mean frequency than the EA-CJ and EA-EJ subcorpora, whereas the latter two did not differ much from each other. Notably, the mean frequency for the abstracts of the empirical research articles was markedly higher than that for the abstracts of the non-empirical academic articles.

A 2 (empirical/non-empirical) × 3 (CA-CJ/EA-CJ/EA-EJ) between-subjects multivariate analysis of variance (MANOVA) was performed to test whether there were significant differences in the use of hedges and boosters between the article types and among the subcorpora. MANOVA was an appropriate statistical test for our data because, as Tabachnick and Fidell (2007) point out, it is preferable to univariate analysis of variance (ANOVA) when the dependent variables are moderately or highly negatively correlated. As discussed earlier, hedges and boosters are metadiscursive resources that can be used to express opposite authorial stances (e.g., uncertainty vs. certainty about a proposition) and/or to modify illocutionary force in opposing directions (see Holmes, 1984; Millan, 2008). Thus, there was good reason to expect the use of hedges and boosters in the same text to be negatively correlated. Of the various MANOVA test statistics, Pillai's trace was chosen because of its robustness (Tabachnick and Fidell, 2007).[3] There was a significant main effect of subcorpus, $F(4, 1286) = 18.28$, $p < .001$, $\eta_p^2 = .05$, indicating that the three subcorpora differed significantly in the mean frequency with which hedges and boosters were used. There was also a significant main effect of article type, $F(2, 642) = 14.57$, $p < .001$, $\eta_p^2 = .04$, with the mean frequency of hedges and boosters differing significantly between the abstracts of the empirical and non-empirical articles. The interaction between subcorpus and article type was not significant, $F(4, 1286) = 1.13$, $p = .34$, $\eta_p^2 < .01$, indicating that the differences among the subcorpora were consistent across the two article types.

The main MANOVA was followed up by two separate 2 × 3 ANOVAs on the use of hedges and boosters, respectively. To control the overall Type I error rate, a Bonferroni correction was applied to the α-level of the ANOVAs (Field, 2009; Tabachnick and Fidell, 2007). The ANOVA on the use of hedges found a significant main effect of subcorpus, $F(2, 643) = 31.92$, $p < .001$, $\eta_p^2 = .09$. Following Field's (2009) recommendation, the Games–Howell test was used as a post hoc procedure to find out where significant between-groups differences lay. The post hoc test revealed that the EA-EJ subcorpus ($M = 2.04$, $SD = 1.99$) used significantly ($p < .001$) more hedges than the CA-CJ subcorpus ($M = 0.84$, $SD = 1.09$) and the EA-CJ subcorpus ($M = 0.98$, $SD = 1.10$), whereas the latter two subcorpora did not differ ($p = .34$). See Fig. 1 for a graphic representation of these patterns. The ANOVA did not find a significant main effect of article type, $F(1, 643) = .50$, $p = .48$, $\eta_p^2 < .01$, indicating that the abstracts of the empirical and non-empirical articles did not differ in their use of hedges. Nor was a significant interaction found between subcorpus and article type, $F(2, 643) = 1.37$, $p = .25$, $\eta_p^2 < .01$.

[2] A Bonferroni correction is an adjustment made to the significance level of a statistical test to prevent the overall Type I error rate from inflating when multiple statistical tests are conducted. It is obtained by dividing the desired global significance level by the number of significance tests conducted. According to Field (2009), it "is a simple but effective correction" (p. 782).

[3] Pillai's trace, also known as the Pillai–Bartlett trace, is one of the commonly used test statistics in multivariate analysis of variance (MANOVA). It is most powerful when the groups compared differ on more than one dimension (Tabachnick and Fidell, 2007); it is also more robust to violations of statistical assumptions than the other test statistics (Field, 2009).

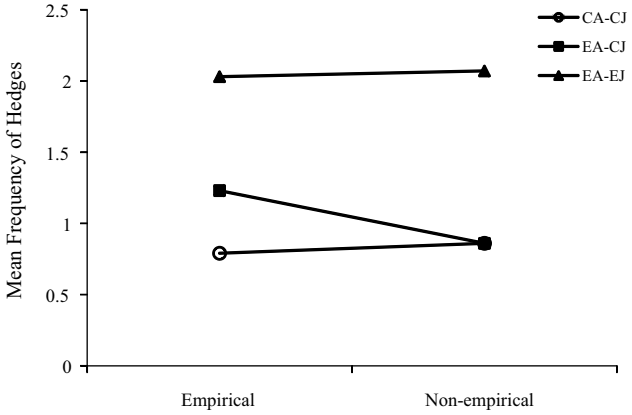


Fig. 1. Mean frequency of hedges by subcorpus and article type.

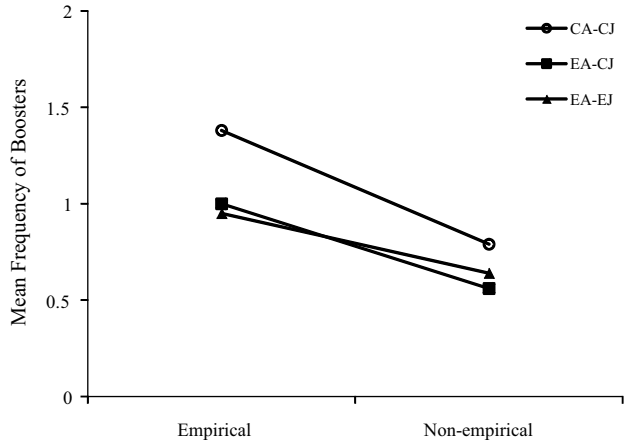


Fig. 2. Mean frequency of boosters by subcorpus and article type.

The ANOVA on the use of boosters found a significant main effect of subcorpus, $F(2, 643) = 6.05$, $p = .002$, $\eta_p^2 = .02$. A post hoc analysis using the Games–Howell test revealed that the CA-CJ subcorpus ($M = 0.98$, $SD = 1.03$) used significantly ($p = .006$) more boosters than the EA-CJ subcorpus ($M = 0.70$, $SD = 0.87$), but did not differ statistically ($p = .14$) from the EA-EJ subcorpus ($M = 0.88$, $SD = 1.05$), whereas the EA-CJ and EA-EJ subcorpora did not differ significantly ($p = .61$) from each other. Fig. 2 illustrates these patterns clearly. The ANOVA also found a significant main effect of article type, $F(1, 643) = 28.67$, $p < .001$, $\eta_p^2 = .04$, indicating that the abstracts of the empirical research articles ($M = 1.07$, $SD = 0.97$) used significantly more boosters than the abstracts of the non-empirical academic articles ($M = 0.67$, $SD = 0.96$). However, no significant interaction was found between subcorpus and article type, $F(2, 643) = .89$, $p = .41$, $\eta_p^2 < .01$.

3.2. Interactive effects of hedges and boosters on authorial stance

Whereas the ANOVAs reported in the previous section analyzed the use of hedges and boosters separately, textual analyses revealed that the patterning of hedges and boosters had a joint interactive effect on the authorial certainty and confidence conveyed in the abstracts. A notable contrast between the CA-CJ and EA-EJ subcorpora emerged from the qualitative analyses. As illustrated below, abstracts in the CA-CJ subcorpus generally conveyed appreciably greater authorial certainty, commitment and assertiveness as a joint function of a markedly lower frequency of hedges and a higher frequency of boosters. By contrast, abstracts in the EA-EJ subcorpus typically expressed much more tentative stances as a result of a much higher incidence of hedges (especially epistemic lexical verbs and modal auxiliaries such as *suggest*, *argue*, *could*, *may* and *might*) and a lower incidence of boosters. Thus, the characteristic patterning of hedges and boosters in the two subcorpora produced distinct authorial voices: whereas authors of the abstracts in the EA-EJ subcorpus appeared more prudent and tentative when interpreting results or presenting claims, authors of the abstracts in the CA-CJ subcorpus sounded more authoritative and confident.

This cross-linguistic contrast was most evident in the abstracts accompanying the empirical articles, as illustrated by the excerpts of two abstracts reproduced below. Representative of the typical frequency and patterning of hedges and boosters

in the EA-EJ subcorpus, Excerpt 1 is the latter part of an abstract for a study designed to investigate the phonemic awareness and lexical processing of foreign language students from Hong Kong and Mainland China. The use of successive hedges (highlighted in italics in Excerpt 1), coupled with the complete absence of boosters, communicates a stance of palpable circumspection in the authors' knowledge claims. Using such hedges as *suggest* and *may*, the authors mark their knowledge claims as provisional and avoid making strong generalizations. In this way, the authors convey their awareness of possible alternative viewpoints held by prospective readers.

Excerpt 1

. . ..The Mainland Chinese IPA-trained participants performed better than both the Hong Kong participants and the Mainland Chinese non-IPA-trained participants in initial phoneme deletion. However, both Mainland Chinese groups outperformed the Hong Kong group on a phoneme-grapheme nonword matching task. This pattern of results *suggests* that phonemic awareness in Chinese L1 readers of English is not simply an effect of orthography, but rather, *may* be interpreted in terms of access to explicit demonstration of phonemes. Further, tests carried out in L2 which are intended to assess metalinguistic awareness *may* be susceptible to artefacts introduced by the participants' L1 spoken language. (*TQ*32, EA-EJ)

By contrast, a converse patterning of hedges and boosters, characteristic of the CA-CJ subcorpus, can be seen in Excerpt 2. This abstract is for an article reporting an empirical study of Chinese learners' acquisition of verb raising in English. The repeated use of boosters (highlighted in bold in Excerpt 2), together with the absence of hedges, contributes to a strong tone of certainty and confidence. The abstract is also representative of the CA-CJ subcorpus in that the author adopts a metadiscursive strategy widely used in the subcorpus – the combination of epistemic lexical verbs (e.g., 表明[*demonstrate*] and 说明[*show*]) with what Hyland (1998b) refers to as "abstract rhetors," namely "research nouns" such as *result* and *finding* or "text nouns" such as *study* and *research* (Charles, 2006). Such metadiscursive collocations allow authors to validate their knowledge claims by signaling that they are factual statements rather than interpretations.

Excerpt 2

本文通过实验探讨中国英语学习者对英语动词提升的习得。实验**表明**，他们知道英语定式句中的题元动词应该位于…。这**说明**受试具有英语定式句中题元动词不能显性提升而情态助动词必须显性提升的句法知识。此结果也**表明**中介语句法系统是没有损伤的，功能语类及其特征值在二语中能够得以体现。(*FLTR*45, CA-CJ)

(This paper examines the acquisition of English verb raising by Chinese learners of English through an experiment. The experiment **demonstrated** that the participants knew that English finite thematic verbs should be placed. . .. These results **showed** that the participants had the correct syntactic knowledge that English finite thematic verbs do not raise, whereas English modal verbs raise. These results also **demonstrated** that the interlanguage syntactic system was not impaired and that functional categories and their parameters were present in L2.)

The contrasting patterning of hedges and boosters illustrated above also extended to the abstracts for the non-empirical articles in the two subcorpora. Excerpt 3 is part of an abstract in the EA-EJ subcorpus that accompanying a critical review of the existing analytical frameworks adopted to analyze second language users' autobiographical narratives. Similarly to Excerpt 1, no boosters are used in this representative excerpt. The author, however, uses two hedges strategically to justify a need for further research and to soften the assertiveness of a major knowledge claim. Using *not always* as a politeness strategy, the author mitigates the illocutionary force of a criticism, a face threatening act (Myers, 1989) to other researchers in the community she belongs to, and creates a knowledge gap to be filled. In a similar vein, the major knowledge claim in the abstract was marked by *I argue* as a personal opinion rather than a conclusion based on firm empirical evidence. The strategic use of hedges, coupled by an absence of boosters, enables the abstract to communicate a tone of palpable tentativeness.

Excerpt 3

In the past decade, language memoirs, linguistic autobiographies, and learners' journals and diaries have become a popular means of data collection in applied linguistics. It is *not always* clear however how one should go about analyzing these data.. . . I *argue* that some analytical approaches, in particular content and thematic analyses, are insensitive to the interpretive nature of autobiographic data.. . . (*AP*8, EA-EJ)

In contrast to the tentativeness conveyed in Excerpt 3, a stance of conviction is expressed in Excerpt 4, part of a typical abstract from the CA-CJ subcorpus. The article in question is a "summing-up report" reflecting on "personal experience and views" (Gao et al., 2001). Although the author hedges a little on the purpose of the article with 尝试(i.e., *attempt*), she uses the booster 表明(i.e., *show*) to strengthen the relationship between available evidence and a wide-ranging claim, assuming a voice of confidence in a firm conclusion and closing off possible alternative views.

Excerpt 4

…本文*尝试*将澳洲ESL一体化教学中的听写策略、声音策略、话题策略引入中国英语写作教学,并对具体策略的实施流程及其特点进行较深入的探讨。实践**表明**,基于澳洲ESL一体化教学模式的英语写作教学策略不仅有助于激发学生的兴趣,开发其创意潜力,而且还能在互动式教学中实现英语写作的开放性和多元化。(*FLW*38, CA-CJ)

(…This paper *attempts* to introduce the dictogloss strategy, the sound sequence strategy and the topic strategy from the Australian integrated ESL model into the Chinese EFL writing classroom. It also discusses the implementation and characteristics of these instructional strategies. Classroom practice **shows** that the writing instruction strategies adapted from the Australian integrated ESL model not only help to arouse students' interest and develop their creativity, but also promote the openness and pluralism of English writing through interactive teaching and learning.)

4. Discussion

Both the quantitative and the textual analyses reported in the previous section identified a manifest cross-cultural/linguistic contrast: the abstracts in the CA-CJ subcorpus tended to express a notably higher degree of certainty, confidence and/or commitment than the abstracts in the EA-EJ subcorpus. This was achieved mainly through a markedly less frequent use of hedges and a somewhat greater use of boosters in the former, as compared with the latter. These results are consistent with those from several cross-linguistic studies of academic discourse (i.e., Kong, 2006; Martín-Martín, 2008; Martín-Martín and Burgess, 2004; Vold, 2006b). A common finding from those studies is that scholars publishing in English-medium academic journals tend to adopt a more tentative stance than their counterparts publishing in some other language. Concurring with other scholars (e.g., Golebiowski, 2002; Taylor and Chen, 1991), we argue that these identified differences are attributable in part to cultural influences. Specifically, we contend, on the basis of support presented below, that the contrast in epistemic stance and the difference in the use of hedges found in our study can be attributed in large measure to the culturally-based rhetorical conventions and styles of persuasion that prevail in the larger sociocultural contexts in which the English- and Chinese-medium academic journals are embedded, respectively.

Because of the dominant Anglo-American presence in the English-medium journals examined in this study, the prevalent rhetorical norms and discursive preferences in these journals are Anglo-American ones (Braine, 2005; Flowerdew, 2001; Swales, 2004). With their historical roots in Socratic and Aristotelian philosophical traditions, Anglo-American cultures value such epistemological practices as questioning one's own as well as others' ideas and beliefs, independently evaluating received knowledge, and engaging in debate and formal argumentation as a canonical form of knowledge construction (Galtung, 1981; Peng and Nisbett, 1999; Tweed and Lehman, 2002). In academic discourse, these cultural practices are likely to find expression in an emphasis on rational reasoning, a preference for reasoned discourse between individuals as a way to approach clarity and truth, the framing of intellectual ideas in adversarial terms, and the anticipation or construction of counterarguments (Peng and Nisbett, 1999; Toulmin, 2003). This proposition is consistent with the finding from recent comparative studies (e.g., Liao and Chen, 2009; Liu, 2005) of first language writing instruction that instructional materials for English composition attach much importance to addressing opposing viewpoints in argumentative writing. It is also supported by the results of a recent survey of English academic style manuals (Bennett, 2009). Bennett reported that one consistent feature of the surveyed style manuals is their caution about making claims: novice writers are consistently advised to avoid overstatements and to use specific rhetorical devices to mitigate their claims. Given this recognized need to address opposing views and the amount of training in anticipating and generating counterarguments (Peng and Nisbett, 1999), it is no surprise that Anglo-American academic writers capitalize on hedging to make their scholarly writing display the appropriate level of circumspection, tentativeness, and commitment so that their positions, arguments or claims are more tenable or palatable to fellow members of a their discourse communities. Furthermore, as Holmes (1984) points out, Anglo-American academic communities tend to be negative-politeness cultures characterized by an "architecture of social distance, asymmetry, and resentment of impositions" (Brown and Levinson, 1987:245) and "a consequent fascination for devices which attenuate negative affective speech acts" (Holmes, 1984:348). Thus it is natural for Anglo-American academics to tap hedging as a negative politeness strategy (Brown and Levinson, 1987), namely, a strategy intended to avoid or minimize impositions on the audience by mitigating face threatening criticisms and other speech acts of a negative affective nature (Holmes, 1982, 1984; Myers, 1989). Once these discursive practices have become the rhetorical norms of the leading English-medium academic journals, scholars from other rhetorical traditions are under pressure to conform to them if they wish to publish in the journals (Swales, 2002).

The Chinese-medium journals, on the other hand, adopt Chinese rhetorical norms since they are situated in prevalent Chinese cultural practices. With their pedigree in Confucian and Taoist traditions (Nisbett et al., 2001; Tweed and Lehman, 2002), Chinese cultural practices are said to be characterized by a deep-seated sociocognitive belief that "verbal debate and argumentation are not meaningful tools for understanding truth and reality" (Peng and Nisbett, 1999:747) and that truth is self-evident without the need for argument (Bodde, 1991). They foster respect for essential, authoritative knowledge (Nakayama, 1984; Tweed and Lehman, 2002), "a focus on particular instances and concrete cases" (Nisbett et al., 2001:294),

and a tendency to rely on prior beliefs and experience in problem solving. Even the sage Confucius (2003:64) declared that "I transmit rather than innovate" (述而不作). Furthermore, he exhorted that "words should convey their point, and leave it at that" (辞达而已矣; p. 189), warning against allowing embellishments and refinements of language to complicate matters needlessly. Shaped by and reflective of such cultural influences, Chinese rhetorical norms tend to encourage the framing of ideas in non-polemical terms, an inclination to "rely less on formal logic and more on experiential knowledge in reasoning" (Nisbett et al., 2001:301; Matalene, 1985), and a tendency to give less attention to possible counterarguments than Anglo-American rhetorical norms do (Liao and Chen, 2009; Liu, 2005). Thus, written discourse is viewed less as an arena for constructing knowledge and approaching truth than as a venue for announcing knowledge and asserting truth (Matalene, 1985). For example, in the Chinese rhetorical tradition, it would be unnecessary for a writer to prove himself/herself to be knowledgeable because by writing the text he/she is assumed to have authority, credibility, and knowledge (Hinkel, 1999). Given these rhetorical practices, Chinese academic writers publishing in Chinese-medium journals are less compelled than their Anglo-American counterparts to hedge their positions or qualify knowledge claims but more likely to perceive a need to assume a tone of certainty so as to convey authority and credibility. In this regard, Yang (1998) reported a similar tendency for greater conviction in Chinese graduate dissertations on language learning and teaching, as compared with those written by English writers.

Apart from the influences of rhetorical practices, divergent epistemological beliefs about science may also have contributed to the cross-cultural/linguistic discrepancies in the use of hedges and the resultant differences in epistemic stance we have observed. The last 50 years have witnessed a growing awareness in Anglo-American social scientific communities of various problems with scientism in general and positivism in particular (Cohen et al., 2007; Denzin and Lincoln, 2005a; Hua, 1995). Scientism is "the idealist belief in the power of science to modify society using the methods, values, and ideas underlying science" (Ouyang, 2003:117). In other words, it is "an idealization of science that reifies and universalizes scientific precepts into cultural dogma" (Lozada, 2003:109). Positivism refers to a philosophical position that regards natural science as the ideal paradigm of human knowledge, advocates the direct application of its methodological procedures to the social sciences, and values law-like generalizations (Cohen et al., 2007). Advances in the Western philosophy of science have questioned the Baconian conception of science as the systematic accumulation of objective knowledge and problematized the ontological, epistemological, and methodological assumptions of positivism (Cohen et al., 2007; Guba and Lincoln, 1985; Kuhn, 1962; Marks, 1983). Pluralism, subjectivity, and the possibility of multiple interpretations are widely recognized as inherent features of scientific research and knowledge construction in Anglo-American intellectual communities (Denzin and Lincoln, 2005a,b). In view of these developments, it can be argued that the growing rejection of scientism and the declining influence of positivism, together with a sober recognition of the immense complexity and elusiveness of the subject of the social sciences (i.e., human behavior and social phenomena), have led Anglo-American researchers to increasingly perceive the need to qualify knowledge claims, withhold full commitment to assertions, and assume a tone of circumspection or tentativeness through the use of metadiscourse strategies such as hedges. Notably, this account is supported by findings from Gillaerts and Van de Velde's (2010) study of abstracts published in the *Journal of Pragmatics* between 1977 and 2007. The researchers found that the use of hedges rose steadily throughout the various decades, whereas there was a notable drop in the occurrence of boosters.

A reverse epistemological trend, however, has been on the rise in China. Because of the "'humiliation' at the hands of Western aggressors" (MacPhail, 2009:45) that China had suffered in the previous 100 years, scientism was enthusiastically embraced by Chinese intellectuals when it was first introduced to China in the early 1900s (Kwok, 1965; Ouyang, 2003). Science was regarded as the ultimate solution to all problems of Chinese society. Commenting on this worship of science, Hu Shi observed that "During the last thirty years or so there is a name [i.e., science] which has acquired an incomparable position of respect in China; no one, whether informed or ignorant, conservative or progressive, dares openly slight or jeer at it" (1923; cited in Hua, 1995:145). Thus, the epistemological belief, characterized as a "Euro-American disease" by Needham (2004:78), has since taken root in Chinese society (Kwok, 1965) and has prevailed in post-Mao China (Greenhalgh, 2008; Hua, 1995; MacPhail, 2009), "placing the development of science and technology in an unprecedented strategic height that concerns the rise or decline of the country and the life or death of the nation" ("Historic Achievements", 2002). The dominance of scientism in contemporary Chinese intellectual thinking has led to a turn to positivism as the ideal paradigm of research in the social sciences. Gao et al. (2001), for example, have noted that while applied linguistics in the West has been shifting from positivism towards constructivism, applied linguistics in China has been moving fast towards positivism. As pointed out in many philosophical and methodological texts (e.g., Cohen et al., 2007; Guba and Lincoln, 1985; Habermas, 1971), the axiomatic assumptions of positivistic research are hostile to epistemic stances that admit of measured uncertainty, guarded commitment, tentativeness, subjectivity, and the possibility of multiple interpretations. These assumptions are necessarily reflected in metadiscursive strategies. As Hyland (2005a) points out, an important rhetorical aspect of positivist epistemology is that individual authority is subordinate to textual authority and that "writers generally seek to disguise both their interpretive responsibilities and their rhetorical identities behind a screen of linguistic objectivity" (p. 147). One way of achieving this is to reduce the use of hedges. Notably, the explanation offered here can account plausibly for both the markedly lower incidence of hedges in the CA-CJ subcorpus and the strong preference therein for epistemic verbs, such as *show* and *demonstrate*, which serve as boosters. These epistemic verbs carry the implication of objectivity and convey authorial certainty or conviction about the probability of an interpretation (Koutsantoni, 2004).

Interestingly, this study has identified another cross-linguistic difference, which is not a cross-cultural one. Specifically, the English abstracts in the Chinese-medium journals used significantly fewer boosters than their matching Chinese

abstracts, but did not differ statistically from the abstracts in the English-medium journals. One explanation for this difference might be that the Chinese applied linguists were aware of the rhetorical norms preferred by the international community of applied linguists and, consequently, tried to accommodate to them by assuming more tentative and measured authorial stances through a less frequent use of boosters. This explanation, however, can be readily ruled out by the pattern of results observed in the use of hedges. The English abstracts in the Chinese-medium journals did not differ statistically in the use of hedges from their matching Chinese abstracts but used significantly fewer hedges than the abstracts in the English-medium journals. If the Chinese applied linguists had attempted to create a tone of circumspection or measured tentativeness in their English abstracts, a direct and more effective way of doing so would have been for them to use more hedges (Crismore et al., 1993; Hyland, 2005a; Vande Kopple, 1985). We believe that a plausible explanation for the markedly lower incidence of both boosters (compared with the Chinese abstracts) and hedges (compared with the abstracts in the English-medium journals) in the English abstracts of the Chinese-medium journals lies in the English proficiency of the Chinese applied linguists. To manipulate and communicate subtleties of authorial stance in a second language requires a highly sophisticated level of proficiency in that language (Holmes, 1982, 1988; Hyland, 2005a). In other words, a skillful use of second language metadiscursive resources such as hedges and boosters is possible only when a writer has moved beyond the basic syntactic structures and general-purpose vocabulary of the target language and mastered more complex syntactic knowledge, richer vocabulary, and greater pragmatic competence. As English is learned and taught as a foreign language in China, many Chinese authors of the English abstracts in the Chinese-medium journals may not have attained a level of English proficiency that would allow them to use English hedging and boosting strategies as competently as they commanded those in their first language. This explanation is consistent with the results of previous studies (e.g., Flowerdew, 1997; Hinkel, 2005; Hyland and Milton, 1997; Ventola, 1997) which found that second language academic texts in general displayed a smaller variety of metadiscourse strategies than first language academic texts and that writers with higher second language proficiency were able to qualify their claims more appropriately than those with lower second language proficiency.

In addition to the aforementioned cross-cultural/linguistic differences, this study has also found that the use of boosters in the abstracts of the empirical research articles was significantly more frequent than that in the abstracts of the non-empirical academic articles. This result is incompatible with previous discussions of how the use of interactional metadiscourse strategies may vary systematically in relation to empirical vs. non-empirical academic articles. As pointed out in the Introduction, it has been proposed that due to differences in the nature of the two broad genres of academic discourse and in the availability of a standardized text format, authors of empirical research articles are expected to be less visible in their writing than authors of non-empirical academic articles (Dahl, 2004; Fløttum, 2003). As Hyland (2005a) suggests, a less frequent use of hedges and boosters is one effective way of minimizing authorial visibility. Taken together, these views would lead one to expect a more frequent use of boosters in the abstracts of the non-empirical academic articles in this study. However, the opposite has been found. We believe that the observed pattern of use of boosters can be plausibly explained in terms of the nature of the supports that claims in empirical vs. non-empirical academic articles are typically based on. As Toulmin (2003) points out, "just how far we are entitled to commit ourselves depends on the strength of the grounds, reasons or evidence at our disposal" (p. 83). Evidence that authors of empirical research articles base their claims on typically consists of primary data, that is, empirical evidence collected systematically by the authors themselves in research designs constructed specially to verify the claims. This type of evidence is likely to be perceived as providing backing that is firm enough to warrant the use of boosters to convey a considerable level of certainty, conviction and committal (Hyland, 2005a). By contrast, evidence drawn on by authors of non-empirical academic articles is made up of a more varied assortment of supports, including secondary sources of data, anecdotal examples, informal observations, theoretical speculations, and so on. Such supports are likely to lead to indicative rather than definitive conclusions and tend to be seen as constituting weaker backing that does not warrant strong conviction or unqualified commitment, hence fewer opportunities to use boosters.

5. Conclusion

This study has revealed several important cross-cultural, cross-linguistic, and genre-related differences in the use of hedges and boosters. First, the abstracts published in the English-medium applied linguistics journals used hedges markedly more frequently than those published in the Chinese-medium counterparts. Second, the Chinese abstracts in the Chinese-medium journals used significantly more boosters than their matching English versions published in the same journals. Third, the abstracts of the empirical research articles used significantly more boosters than those of the non-empirical academic articles. Fourth, the distinct patterning of hedges and boosters in the Chinese and English abstracts had a joint, interactive effect on the authorial certainty and confidence conveyed therein. These salient differences have been attributed to culturally preferred rhetorical strategies, epistemological beliefs, lack of facility in English as a second/foreign language, and the nature of supporting evidence typically drawn on for knowledge claims in different types of academic writing. These interesting findings notwithstanding, several issues need to be addressed in future studies. To begin with, this study has examined abstracts of journal articles in a single discipline so as to tease apart cultural/language effects on the use of boosters and hedges from disciplinary ones. To develop a more comprehensive understanding of the various influences on the use of these metadiscursive resources, however, it is also necessary to examine, in future research, comparable data from different disciplines (see Lindeberg, 2004) to identify how variations in the disciplinary norms of metadiscursive practices may shape the use of interactional metadiscourse independently and in interaction with cultural/linguistic influences.

Second, this study has focused on only two types of interactional metadiscourse strategies in research article abstracts. Further investigations will need to broaden the scope to include the use of interactive as well as other interactional metadiscursive resources (see Hyland, 2005a) in research article abstracts. Last but not least, future studies should address the lack in this study of rich qualitative data (e.g., extended interviews of Anglo-American and Chinese applied linguists) which would allow a more in-depth understanding of the considerations academic writers take into account when they use hedges and boosters to make their knowledge claims. Research along these interesting avenues can contribute to building up the adequate knowledge base that is required to advance cross-cultural/linguistic scientific communication and support the development of effective second language academic writing programs.

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